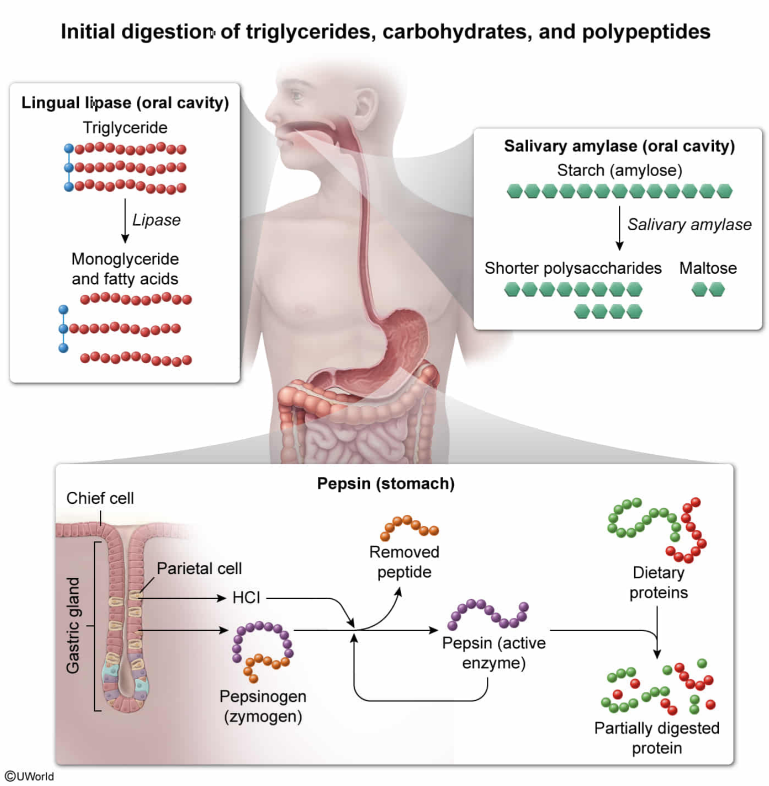
Initial triglyceride, carbohydrate, and polypeptide digestion is performed by:

- A. lingual lipase, salivary amylase, and trypsinogen, respectively. [13%]
- B. bile, pancreatic amylase, and pepsin, respectively. [14%]
- C. bile, pancreatic amylase, and trypsinogen, respectively. [6%]
- ✔ D. lingual lipase, salivary amylase, and pepsin, respectively. [66%]

Correct
65% answered correctly

## Explanation:




The components of the gastrointestinal tract include the oral cavity, esophagus, stomach, small intestine (SI), and large intestine.

Initial mechanical digestion of food occurs via mastication by the teeth in the oral cavity and, at the same time, saliva and mucus secreted by the salivary glands lubricate ingested food. Saliva, a fluid substance, contains enzymes for initial chemical digestion of some macromolecules. These enzymes include **lingual lipase**, which hydrolyzes triglycerides into free fatty acids and diglycerides, and **salivary amylase**, which hydrolyzes the polysaccharide starch into the disaccharide maltose. Further chemical digestion of these macromolecules occurs in the SI.

The muscles of the oral cavity compress mechanically digested food into a bolus (ball), which can be swallowed. This bolus

passes from the oral cavity to the pharynx (throat) and past the upper esophageal sphincter. Next, the bolus enters the esophagus, a passageway for food to be carried to the stomach. Esophageal peristalsis propels the bolus in the direction of the stomach.

In the stomach, gastric chief cells release the zymogen (inactive) pepsinogen, which is converted to its active form, **pepsin**, upon mixing with hydrochloric acid in gastric juice. Pepsin is a proteolytic enzyme responsible for the initial digestion of polypeptides into smaller peptides. Accordingly, initial triglyceride, carbohydrate, and polypeptide digestion is performed by **lingual lipase, salivary amylase, and pepsin**, respectively.

The pancreas secretes the enzymes trypsinogen and pancreatic amylase, which are released into the SI duodenum through the pancreatic duct. Another duodenum enzyme, enteropeptidase, converts trypsinogen to its active form, trypsin. Trypsin activates other pancreatic zymogens (eg, chymotrypsinogen) and functions in continued peptide digestion. However, pepsin, *not* trypsinogen, is responsible for *initial* polypeptide digestion **(Choice A)**. Also in the duodenum, pancreatic amylase facilitates polysaccharide hydrolysis, forming disaccharides. However, salivary (*not* pancreatic) amylase is responsible for *initial* carbohydrate digestion **(Choice B)**.

Bile salts released by the gallbladder into the duodenum break down lipids into smaller droplets during emulsification. This process produces micelles, or spherical structures containing a hydrophobic core and outer polar region, that increase lipid surface area for hydrolysis by lipases. However, *initial* triglyceride digestion occurs via salivary lipase (*not* bile) **(Choice C)**.

**Educational objective:**
The oral cavity aids in ingestion and initial digestion of carbohydrates and triglycerides via salivary amylase and lingual lipase, respectively. A compacted food bolus passes to the esophagus, which carries the bolus to the stomach, where initial digestion of polypeptides by pepsin occurs.

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| **Time spent:** 0 sec | **QID:** 403415 |
| **Subject:** Biology | **System:** Digestion and Excretion |